

Random Enough: A Connectome Test of the Fly Hashing Algorithm

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Abstract—The *Drosophila* olfactory circuit has been characterised as a locality-sensitive hash: a sparse expansion from ~ 50 glomeruli onto ~ 2000 Kenyon cells followed by winner-take-all sparsification, which outperforms classical locality-sensitive hashing (LSH) at similarity search. That result was obtained with a *randomly generated* projection matrix, because no connectome of the circuit existed. The MaleCNS v1.0 connectome makes the assumption testable. We substitute the measured glomerulus-to-Kenyon-cell matrix for the random one and evaluate nearest-neighbour retrieval on measured odorant responses. The architecture reproduces its advantage over classical LSH ($4.8\times$). The measured wiring, however, confers no advantage over a configuration-model null that preserves both degree sequences and randomises only the pairing: $z \in [-1.59, -1.12]$, with 35–38 of 40 null draws scoring higher. Calibrating against matrices of known quality bounds any true advantage below 1.8% relative. This is not a claim that the wiring is unstructured; independent work has shown that it is not. It is the narrower and, we argue, more informative claim that the structure present is not structure that helps this computation.

Index Terms—connectomics, locality-sensitive hashing, similarity search, neural architectures, null models

I. INTRODUCTION

Similarity search — given a query, return the database items most like it — is a primitive of information retrieval, and locality-sensitive hashing (LSH) is its standard sublinear solution [1], [2]. An LSH maps items to short codes such that similar items collide with high probability, replacing expensive comparisons in the input space with cheap ones over codes.

Dasgupta, Stevens and Navlakha [3] observed that the fruit fly’s olfactory circuit computes something structurally similar, and departs from classical LSH in three ways: the projection is *sparse* and *binary* rather than dense and Gaussian; it *expands* rather than reduces dimensionality; and it sparsifies the result by winner-take-all. On standard benchmarks, their “fly algorithm” outperformed classical LSH by up to $3\times$.

Their model of the projection was necessarily synthetic. The glomerulus-to-Kenyon-cell connectivity had not been mapped, so they drew a sparse binary random matrix whose only biological anchor was that each Kenyon cell samples roughly six inputs, matching anatomical estimates. The prevailing view at the time held that this connectivity is effectively random [4], which made the choice defensible; whether it was *accurate* was untestable.

Two developments make it testable. First, Zheng *et al.* [5] used the hemibrain volume to show that Kenyon cell sampling of olfactory input is measurably *not* random, overturning the earlier view. Second, the MaleCNS v1.0 connectome [6]

provides a complete reconstruction of an adult central nervous system, including the full mushroom body input layer.

These two facts create a question that neither answers alone. The wiring is structured; the circuit is claimed to implement a hash. *Does the structure help the hash?* We find that it does not, to within a 1.8% resolution bound, while the architecture’s advantage over classical LSH reproduces. Our contributions are:

- the first evaluation of the fly hashing algorithm using measured rather than synthetic connectivity;
- a null-model protocol that isolates wiring *pairing* from the degree sequences it is confounded with, together with calibration matrices that establish what effect size the protocol can resolve;
- the finding that measured structure yields no similarity-search advantage, which retroactively validates the modelling assumption in [3] while constraining the functional interpretation of the structure reported in [5].

II. BACKGROUND

A. Locality-Sensitive Hashing

A family $\mathcal{H}$ of hash functions over a metric space $(\mathcal{X}, D)$ is (r, cr, p_1, p_2) -sensitive if, for h drawn uniformly from $\mathcal{H}$,

$$D(u, v) \leq r \Rightarrow \Pr[h(u) = h(v)] \geq p_1, \quad (1)$$

$$D(u, v) \geq cr \Rightarrow \Pr[h(u) = h(v)] \leq p_2, \quad (2)$$

with $c > 1$ and $p_1 > p_2$ [1]. Classical constructions reduce dimensionality with dense random projections and binarise, e.g. the sign of a Gaussian projection [2].

B. The Fly Circuit as a Hash

Olfactory receptor neurons converge onto $d \approx 51$ glomeruli, whose projection neurons diverge onto $m \approx 2000$ Kenyon cells. Two mechanisms shape the code. Divisive normalisation in the antennal lobe removes concentration, retaining odour identity [7]. A single giant inhibitory neuron, APL, implements approximate winner-take-all, leaving a few percent of Kenyon cells active [8], [9]. The resulting sparse binary pattern is the circuit’s tag for an odour. The same construction also supports novelty detection [10].

C. Null Models for Connectomes

Asking whether a measured network is “special” requires a null preserving everything uninteresting. Degree-preserving randomisation [11] is standard: it retains each node’s connection count while destroying which node connects to which.

In a bipartite projection, two degree sequences may be preserved independently, and which one is held fixed changes the question being asked.

III. METHODS

A. The Measured Projection

Let $M \in \{0,1\}^{m \times d}$ with $M_{ij} = 1$ iff glomerulus j synapses onto Kenyon cell i . Define the *claw count* of cell i and the *fan-out* of glomerulus j as

$$c_i = \sum_j M_{ij}, \quad f_j = \sum_i M_{ij}. \quad (3)$$

We build M from MaleCNS v1.0 [6] using every reconstructed connection, imposing no synapse-count threshold: the resulting graph has 2.56×10^7 connections and 1.24×10^8 synapses. Thresholding is common when connectomes drive dynamical models [12], but Kenyon cell claws are individually weak, and a threshold removes them: at a ≥ 5 synapse cut the mean claw count falls to 4.79, against 5.28 with no cut and an anatomical expectation near six.

Projection neurons are identified by the ALPN class and assigned to glomeruli from their cell-type labels; multiglomerular and unnamed types are excluded, as are thermo- and hygro-sensory channels. Restricted to one hemisphere this yields $d = 51$ glomeruli and $m = 1886$ Kenyon cells.

B. The Hash

Following [3], for an input $x \in \mathbb{R}_{\geq 0}^d$ of glomerular responses:

$$\tilde{x} = \frac{d}{\sum_j x_j} x \quad (\text{divisive normalisation}), \quad (4)$$

$$y = M\tilde{x} \quad (\text{sparse binary expansion}), \quad (5)$$

$$h_k(x)_i = \mathbf{1}[y_i \geq y_{(k)}] \quad (\text{winner-take-all}), \quad (6)$$

where $y_{(k)}$ is the k -th largest entry of y . The tag $h_k(x) \in \{0,1\}^m$ has exactly k ones. Tags are compared by Hamming distance.

The classical baseline draws $W \in \mathbb{R}^{d \times k}$ with i.i.d. Gaussian entries and returns $\text{sign}(\tilde{x}^\top W)$, so both methods expend k bits.

C. Null and Calibration Models

Only M varies; input, normalisation, sparsity and scoring are held fixed.

- $\mathcal{N}_{\text{claw}}$: preserves $\{c_i\}$; each cell's inputs are drawn uniformly without replacement. Fan-out is flattened.
- $\mathcal{N}_{\text{conf}}$: a configuration model preserving *both* $\{c_i\}$ and $\{f_j\}$, randomising only the pairing. This is the null of record: any difference it exposes cannot be attributed to either degree sequence.
- $\mathcal{N}_6$: the construction of [3], six uniformly drawn inputs per cell.

Two further matrices are not nulls but calibration probes, used to establish what the protocol can detect: $\mathcal{P}_{\text{bal}}$, which preserves $\{c_i\}$ while making fan-out as uniform as possible, and $\mathcal{P}_{\text{deg}}$, in which every cell draws from the same four glomeruli.

TABLE I
MEAN AVERAGE PRECISION AT $\kappa = 10$, $n = 4000$, $B = 40$.

k	real	$\mathcal{N}_{\text{claw}}$	$\mathcal{N}_{\text{conf}}$	$\mathcal{N}_6$	LSH
8	0.1965	0.2102	$0.2110 \pm .0090$	0.2469	0.0182
16	0.2882	0.3015	$0.2966 \pm .0060$	0.3065	0.0662
32	0.3531	0.3699	$0.3593 \pm .0042$	0.3535	0.1514
64	0.3960	0.4137	$0.4000 \pm .0036$	0.3847	0.2462

D. Evaluation

For query q let $\text{NN}_\kappa(q)$ be its κ nearest items by Euclidean distance in $\mathbb{R}^d$ and $R_\kappa(q)$ the κ retrieved by Hamming distance on tags, ranked. With $\text{rel}(i) = \mathbf{1}[R_\kappa(q)_i \in \text{NN}_\kappa(q)]$,

$$\text{AP}_\kappa(q) = \frac{1}{\kappa} \sum_{i=1}^{\kappa} \text{rel}(i) \cdot \frac{\sum_{l \leq i} \text{rel}(l)}{i}, \quad (7)$$

and mAP averages over queries. We use $\kappa = 10$.

Writing $s(M)$ for the mAP achieved by matrix M , and μ, σ for the mean and standard deviation of s over $B = 40$ draws from $\mathcal{N}_{\text{conf}}$, we report

$$z = \frac{s(M_{\text{real}}) - \mu}{\sigma}, \quad \delta_{2\sigma} = \frac{2\sigma}{\mu}. \quad (8)$$

Here $\delta_{2\sigma}$ is the smallest relative advantage the run could resolve at two standard deviations; a true advantage must be smaller than it. Reporting $\delta_{2\sigma}$ converts an unfalsifiable negative into a bound.

E. Data

Odorant responses come from DoOR 2.0 [13], mapped from receptor to glomerulus. Its consensus matrix is $\sim 17\%$ populated, so we retain odorants measured across at least 12 glomeruli and glomeruli measured across at least 40 odorants, giving 172 odorants over 35 glomeruli; the projection is restricted accordingly to 35×1838 . Because 172 items leave the null too broad to resolve small effects, queries are Dirichlet-weighted blends of two to five measured odorants — mixtures being what a fly actually encounters — yielding $n = 4000$ items that retain the natural correlation structure.

IV. RESULTS

A. The Architecture Reproduces Its Advantage

Table I and Fig. 1 (left) show the expansion plus winner-take-all construction outperforming the Gaussian baseline at equal bit budget by $4.8\times$ on average, qualitatively reproducing [3]. We stress that this is a reproduction in kind, not in magnitude: the measured matrix has fixed dimensions, so the benchmarks of [3] cannot be passed through it, and our evaluation is at the circuit's own scale on its own input modality.

B. The Measured Wiring Confers No Advantage

Against $\mathcal{N}_{\text{conf}}$ the measured matrix yields $z = -1.59, -1.39, -1.47, -1.12$ for $k = 8, 16, 32, 64$, with 38, 35, 37, 36 of 40 null draws scoring higher. The point estimate is consistently *below* the null, though never significantly so.

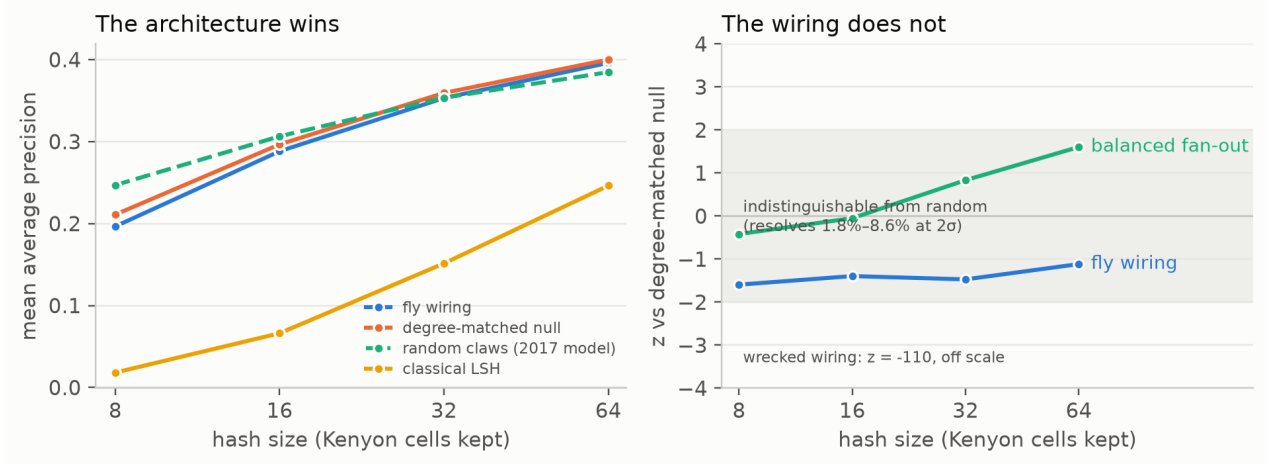


Fig. 1. Left: the architecture’s advantage over classical LSH reproduces with measured wiring; the fly traces sit together well above the LSH trace. Right: against a configuration-model null the measured wiring lies inside the band of indistinguishability at every hash size, while a wrecked matrix is rejected at $z < -22$ (off scale).

TABLE II
WHAT THE PROTOCOL CAN RESOLVE.

k	$\delta_{2\sigma}$	$z(\mathcal{P}_{\text{bal}})$	$z(\mathcal{P}_{\text{deg}})$
8	8.6%	-0.42	-22.9
16	4.1%	-0.05	-48.6
32	2.4%	+0.83	-83.8
64	1.8%	+1.61	-110.0

C. Calibration and Resolution

Table II shows the protocol rejecting a degraded matrix overwhelmingly. An *improvement* of the size produced by $\mathcal{P}_{\text{bal}}$ approaches but does not cross the threshold, which is why we state a bound rather than an absence: any true advantage of the measured wiring is below 1.8% relative at $k = 64$.

D. Robustness

The conclusion is unchanged under a fourfold change in graph density: at a ≥ 5 synapse threshold (6.24×10^6 connections) we obtain $z \in [-1.82, -1.12]$, against $[-1.59, -1.12]$ with no threshold. It also holds for the contralateral hemisphere, for synapse-weighted rather than binary M , and for Gaussian, low-rank and sparse synthetic inputs, with z ranging over $[-0.81, +1.25]$ and no consistent sign.

One asymmetry is worth recording. Measured fan-out is strongly non-uniform: DA1, the pheromone glomerulus, drives 473 Kenyon cells while the quietest drives 19. Both $\mathcal{N}_{\text{claw}}$ and $\mathcal{P}_{\text{bal}}$, which flatten this, score *above* the measured matrix at every k . To the extent that the fly allocates disproportionate mushroom body real estate to behaviourally privileged channels, that allocation appears to cost general-purpose retrieval performance.

V. DISCUSSION

Two readings must be kept apart. Our result is *not* that glomerulus-to-Kenyon-cell wiring is random; Zheng *et al.*

[5] demonstrated otherwise, and our own fan-out asymmetry is itself structure. The result is that the structure present confers no measurable advantage on similarity search — the computation the circuit has been argued to implement.

This resolves rather than contradicts. The modelling assumption in [3] was not merely convenient but accurate *for its purpose*: with respect to retrieval quality, the measured matrix and a degree-matched random one are interchangeable. Where a computation calls for a random projection, evolution appears to have supplied one, and the deviations from randomness that exist serve something else — plausibly innate valence, developmental constraint, or the over-allocation to privileged channels visible in the fan-out.

More broadly, the availability of complete connectomes invites a class of experiment that was previously impossible: taking a circuit-derived algorithm and asking whether the measured substrate outperforms its idealisation. Our protocol — fix everything but the wiring, null out the degree sequences separately, and calibrate what the test can see — is not specific to olfaction and transfers to any connectome-grounded computational claim.

VI. LIMITATIONS

The claim is a bound, not an absence: advantages below 1.8% relative are not excluded, and tightening it requires more items, which costs $O(n^2)$ in the retrieval evaluation. We test one task and one metric; novelty detection [10], the circuit’s other documented function, is untested. DoOR covers 35 of 51 glomeruli, and the 4000 queries are mixtures of 172 measured odorants rather than independent measurements. MaleCNS is a single male animal; whether the conclusion holds across individuals, or against the female FlyWire reconstruction [14], is open.

VII. CONCLUSION

Substituting measured connectivity for the random matrix assumed in [3] leaves similarity-search performance un-

changed to within 1.8%, while the architectural advantage over classical LSH reproduces. The mushroom body is a good hash function because of its shape, not its wiring. Code, data pipeline and results are available at <https://github.com/ssenge/FlyHash-Connectome>.

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